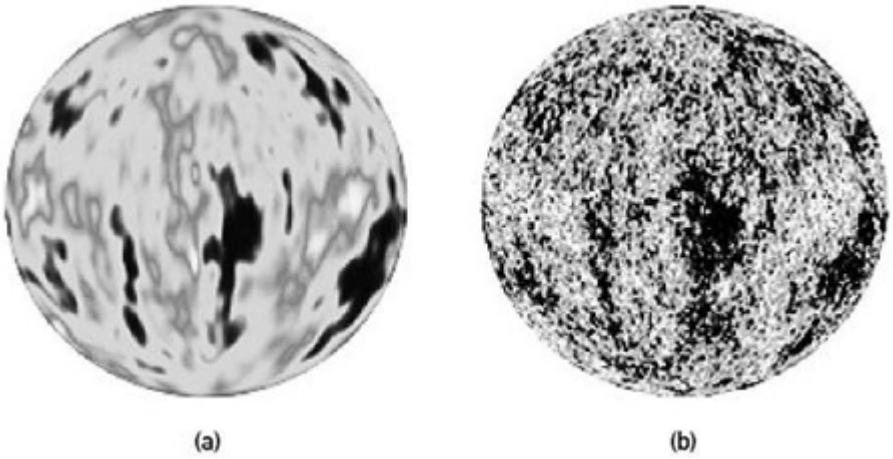


Figure 14.4 (a) Cosmic microwave background radiation data gathered by the COBE satellite. The radiation has been traveling through space unimpeded since about 300,000 years after the big bang, so this picture renders the tiny temperature variations present in the universe nearly 14 billion years ago. (b) Improved data collected by the WMAP satellite.

The influx of precision data has winnowed the field of cosmological proposals, with the inflationary model being, far and away, the leading contender. But as we mentioned in Chapter 10, inflationary cosmology is not a unique theory. Theorists have proposed *many* different versions (old inflation, new inflation, warm inflation, hybrid inflation, hyperinflation, assisted inflation, eternal inflation, extended inflation, chaotic inflation, double inflation, weak-scale inflation, hypernatural inflation, to name just a few), each involving the hallmark brief burst of rapid expansion, but all differing in detail (in the number of fields and their potential energy shapes, in which fields get perched on plateaus, and so on). These differences give rise to slightly different predictions for the properties of the microwave background radiation (different fields with different energies have slightly different quantum fluctuations). Comparison with the WMAP and Planck data should be able to rule out many proposals, substantially refining our understanding.

In fact, the data may be able to thin the field even further. Although quantum fluctuations stretched by inflationary expansion provide a compelling explanation for the observed temperature variations, this model has a competitor. The cyclic cosmological model of Steinhardt and Turok, described in Chapter 13, offers an alternative proposal. As the two three-branes of the cyclic model slowly head toward each other,